

Chapter 1 Riemannian Metrics

1.1 Definitions

Definition 1.1.0.1

- Let M be a smooth manifold. A **Riemannian metric** on M is a 2-tensor field $g \in \mathcal{T}^2 M$, whose value

$$g_p : T_p M \times T_p M \rightarrow \mathbb{R}$$

which is positive definite in the sense of $g_p(v, v) \geq 0$.

- A **Riemannian manifold** is a pair (M, g) .

Remark Riemannian metric vs Metric Space

♣ **Example 1.1.0.2** (Euclidean Metric)

Isometries

Local Representation

Definition 1.1.0.3 (Local Representation of Riemannian Metric)

♠ **Proposition 1.1.0.4** (Existence of Orthonormal Frame)

◦ **Proof ...**

□

Definition 1.1.0.5 (Unit Tangent Bundle) Define the **unit tangent bundles**

$$UTM := \{(p, v) \mid |v|_g = 1\}$$

♠ **Proposition 1.1.0.6** If (M, g) is a Riemannian-Manifold with or without boundary, its unit tangent bundle UTM is a smooth, properly embedded codimensional-1 submanifold with boundary in TM , with

$$\pi : UTM \rightarrow M \quad \partial(UTM) = \pi^{-1}(\partial M)$$

Then, UTM is connected $\Leftrightarrow M$ is connected. UTM is compact $\Leftrightarrow M$ is compact.

◦ **Proof ...**

□

1.1.1 Riemannian Submanifold

◇ **Lemma 1.1.1.1** Suppose $(\tilde{M}, \tilde{g})$ is a Riemannian manifold with or without boundary. M is a smooth manifold with or without boundary. And

$$F : M \rightarrow \tilde{M}$$

is a smooth map. The smooth 2-tensor field (pullback)

$$g = F^* \tilde{g}$$

is a Riemannian metric iff F is an immersion.

◦ **Proof ...**

□

1.2 Models of Riemannian Manifolds

1.2.1 Symmetries

1.2.2 Covering Space of Riemannian Manifold

Euclidean Space

Sphere

Hyperbolic Space

Killing-Hopf Theorem

► **Theorem** Suppose (M, g) be complete, simply connected Riemannian n -manifold with constant sectional curvature, $n \geq 2$.

Then, M is isometric to one of the model spaces $\mathbb{R}^n, \mathbb{S}^n(R), \mathbb{H}^n(R)$.

◦ **Proof ...**

□

★ **Corollary 1.2.2.1 ...**

◦ **Proof ...**

□

Space Form

Chapter 2 Riemannian Connections

2.1 Affine Connections

Definition 2.1.0.1 Define the **affine connections**

$$\begin{aligned} \nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) &\rightarrow \mathfrak{X}(M) \\ (X, Y) &\mapsto \end{aligned}$$

With following properties:

- $(C^\infty(M)\text{-Linear}) \nabla_{fX+gY}Z = f\nabla_XZ + g\nabla_YZ;$
- $(\mathbb{R}\text{-Linear}) \nabla_X(aY + bZ) = a\nabla_X(Y) + b\nabla_X(Z)$
- $(\text{Lebnitz}) \nabla_X(fY) = f\nabla_XY + X(f)Y$

where $X, Y, Z \in \mathfrak{X}(M), f, g \in C^\infty(M)$.

♠ **Proposition 2.1.0.2** Let M be a differentiable manifold with an affine connection ∇ . There exists a unique correspondence which associates to a vector field V along the differentiable curve

$$c : I \rightarrow M$$

another vector field along c , called the **covariant derivative** of V along c , such that:

-
-
-

◦ **Proof ...**

□

2.1.1 Parallel Vector Field

2.2 Riemannian Metrics, Levi-Civita Theorems

Chapter 3 Geodesics

3.1 The Geodesics Flow

3.2 Minimalizing Properties

Chapter 4 Curvatures

4.1 Curvatures

4.2 Sectional Curvature

4.3 Ricci Curvature and Scalar

4.4 Tensors on Riemannian Manifolds

4.5 Manifolds of Nonpositive Curvature

4.5.1 Cartan-Hadamard Manifold

► *Theorem* (Cartan-Hadamard)

◦ Proof ... □

Definition 4.5.1.1 ...

♠ Proposition 4.5.1.2 ...

◦ Proof ... □

♠ Proposition 4.5.1.3 ...

◦ Proof ... □

4.5.2 Cartan's Fixed Point Theorem

4.5.3 Cartan's Torsion Theorem

Chapter 5 Riemannian Submanifold

Chapter 6 Gauss-Bonnet Theorem

Chapter 7 Jacobi Field

Chapter 8 Comparison Theory

Chapter 9 Curvature and Topology

Chapter 10 Isomorphic Immersions

Chapter 11 Complete Manifold, Hopf-Rinow and Hadamard Theorems

Chapter 12 Spaces of Constant Curvature

Chapter 13 Variations of Energy

Chapter 14 The Morse Index Theorem

Chapter 15 The Fundamental Group of Manifolds of Negative Curvatures

Chapter 16 The Sphere Theorem

Chapter 17 The Laplacian

This will be given in geometric analysis.